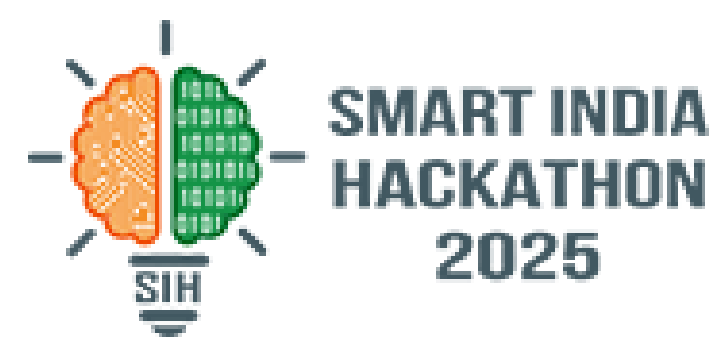
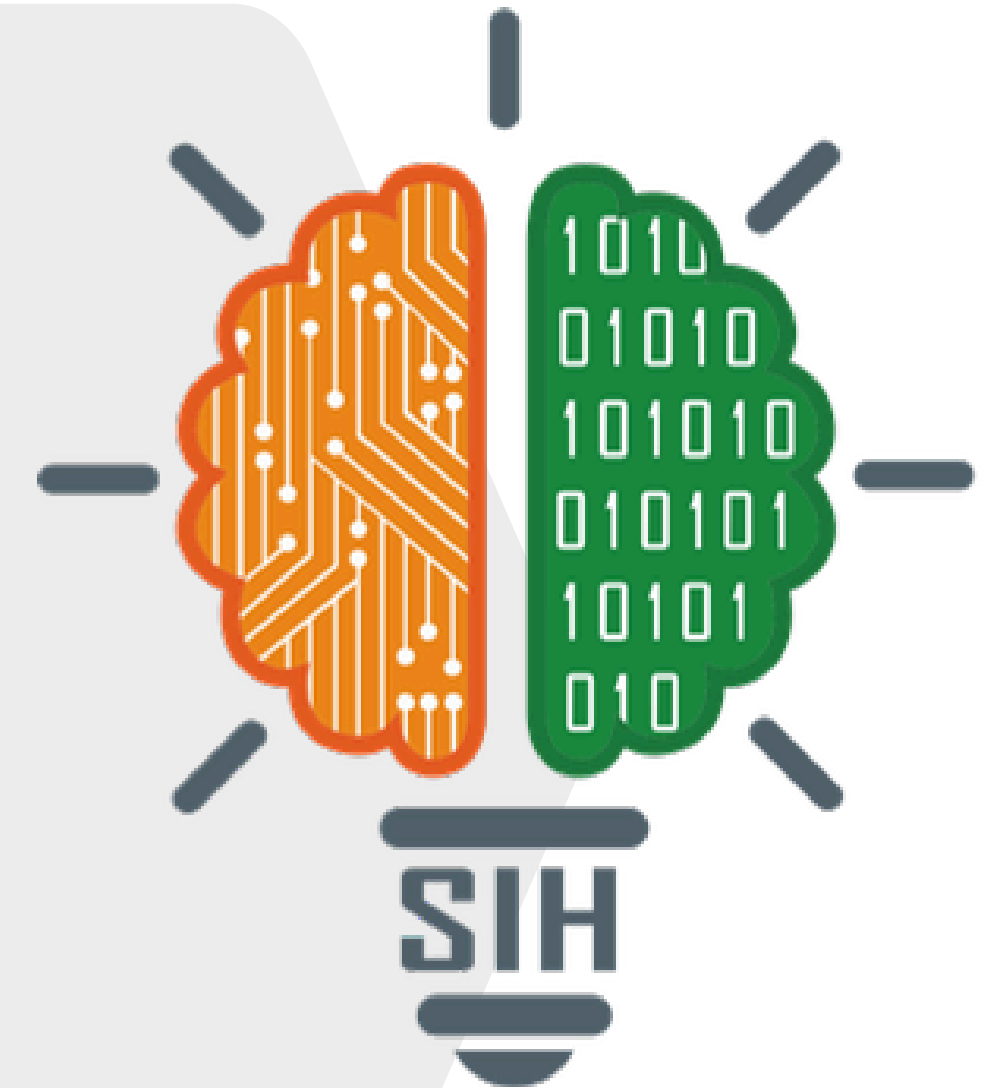


SMART INDIA HACKATHON 2025



- **Problem Statement ID – 25050**
- **Problem Statement Title- SMART TRAFFIC MANAGEMENT SYSTEM
FOR URBAN CONGESTION**
- **Theme- TRANSPORTATION & LOGISTICS**
- **PS Category- Software**
- **Team ID- 92770**
- **Team Name (Registered on portal) - 404 The Optimists**

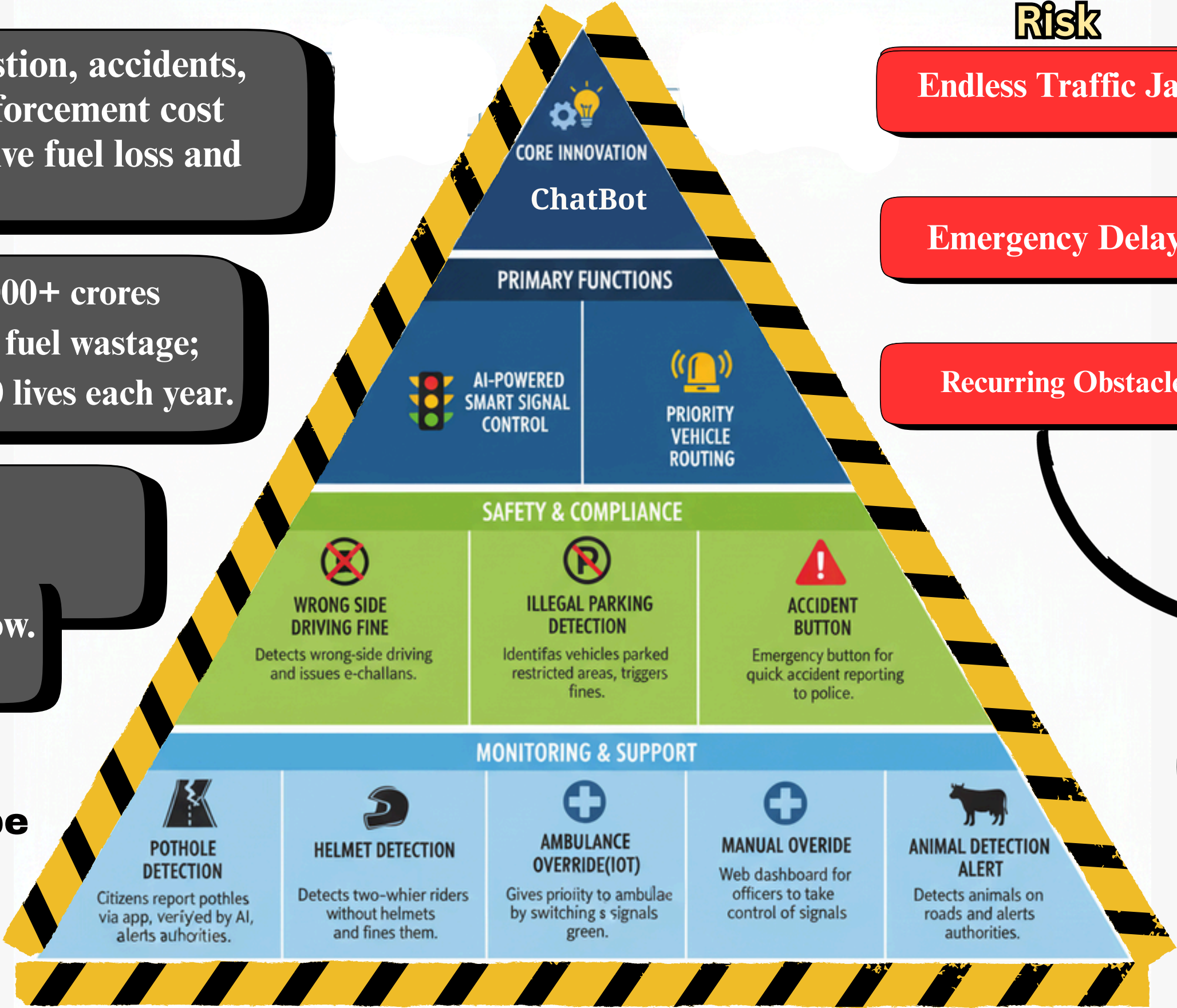


Lanezy: AI-Powered Adaptive Traffic Control Revolutionizing India's Roads

Real-world issue: Urban congestion, accidents, emergency delays, and poor enforcement cost billions annually, causing massive fuel loss and significant time loss.

Why important: India loses ₹60,000+ crores annually due to traffic delays and fuel wastage; road accidents claim over 150,000 lives each year.

Solution: AI system optimizes signals, detects anomalies, ensuring safer, faster traffic flow.



Risk

vs

Solution

Endless Traffic Jams

Smart Signal Control

Emergency Delays

Priority Vehicle Routing

Recurring Obstacles

Real-time Detection & Alerts

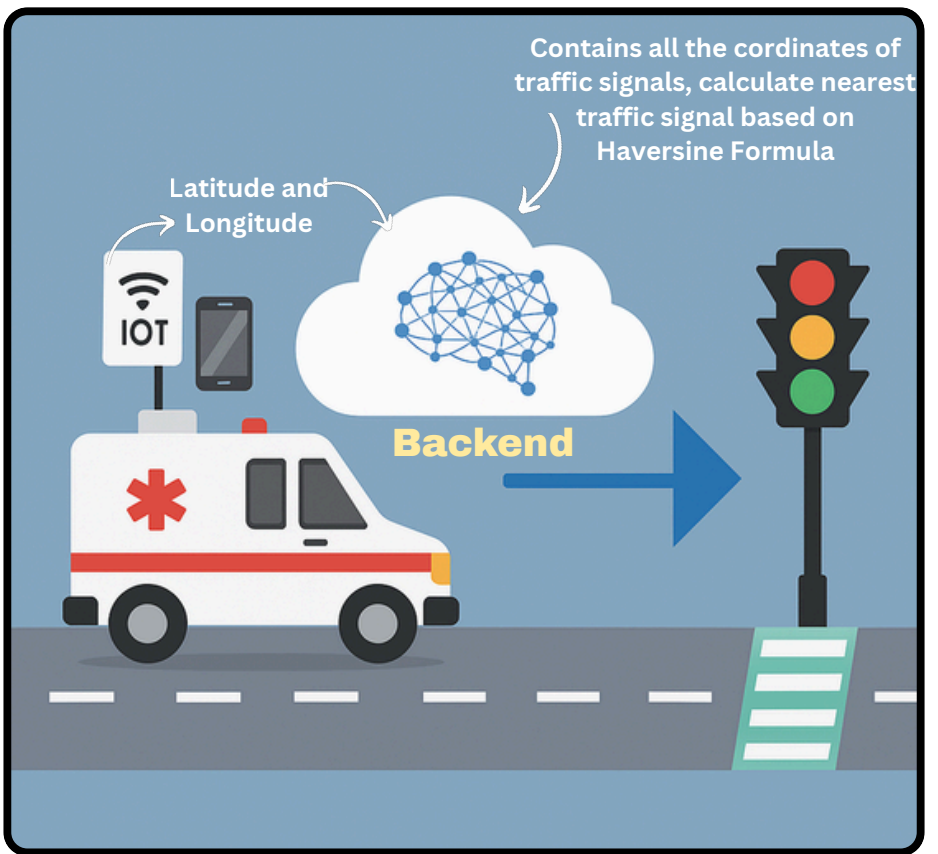
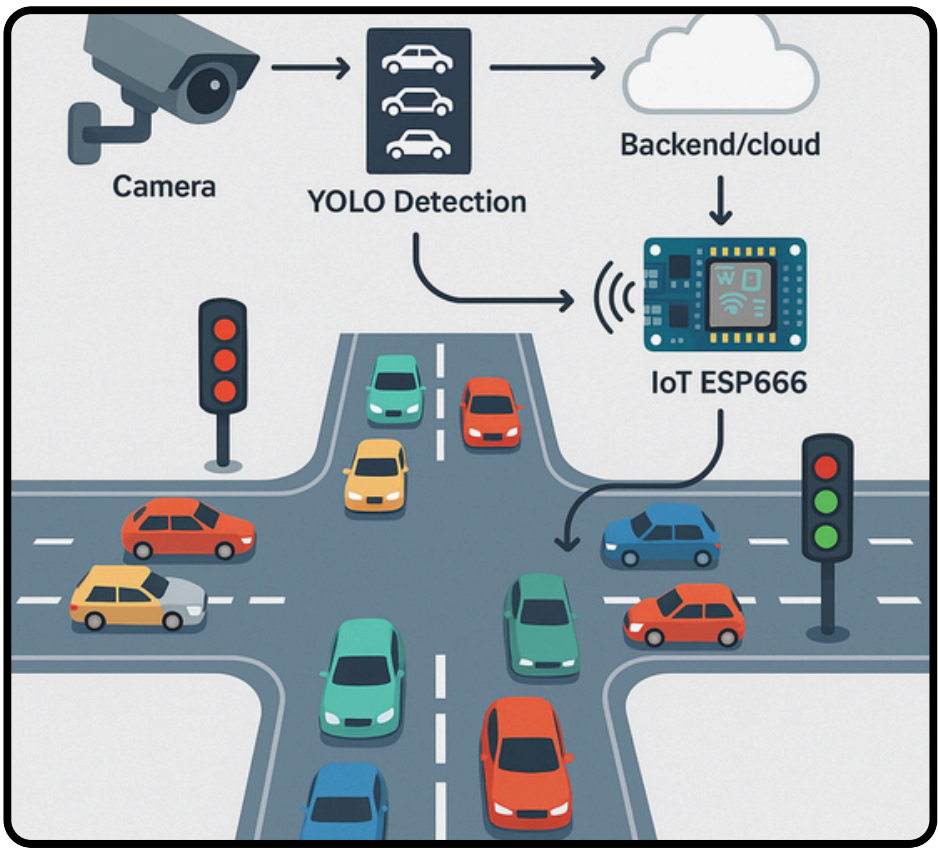
Prototype

[Video](#)

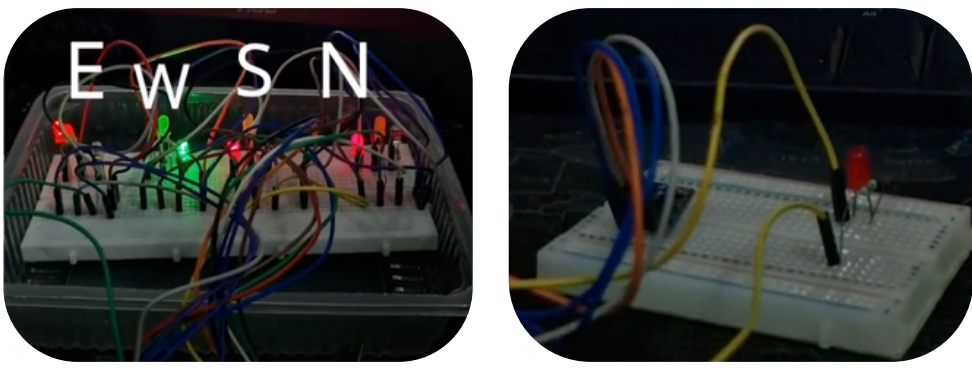
[Website](#)



METHODOLOGY & PROCESS OF IMPLEMENTATION



SYSTEM IOT



TECHNOLOGIES USED

**FrontEnd**
React and TailwindCSS are used for the frontend development.

**BackEnd**
FastAPI is used for the backend development.

**Deployment**
Render and Vercel are used for deployment.

**Hardware**
LEDs, GPS Module, ESP8266, YOLOv8.pt, QMC Module, and Resistors are used.

FEASIBILITY ANALYSIS



Infrastructure Reusability: Leverages existing traffic signals and cameras, requiring minimal additional hardware installation



Scalable Deployment: Can be implemented at a few key intersections initially and gradually expanded citywide as needed.



Cost-Effectiveness: Setup cost is relatively low compared to building entirely new infrastructure.



Authority Integration: Includes features like SOS alerts, obstacle detection, and manual override for seamless police control.

VIABILITY



Proven Concept: Leverages the success of existing AI-driven traffic systems in major cities, demonstrating clear market demand and viability.



Builds Public Trust: Automated detection minimizes human bias, while timestamped video evidence provides credible, transparent proof for all incidents.



Guarantees Accountability: Fosters transparency through detailed logs, acknowledgements, and performance analytics, ensuring every action is traceable.



Ensures Authority Engagement: Real-time dashboards and one-click dispatch alerts keep operators actively engaged and enable rapid response.

BUSINESS POTENTIAL

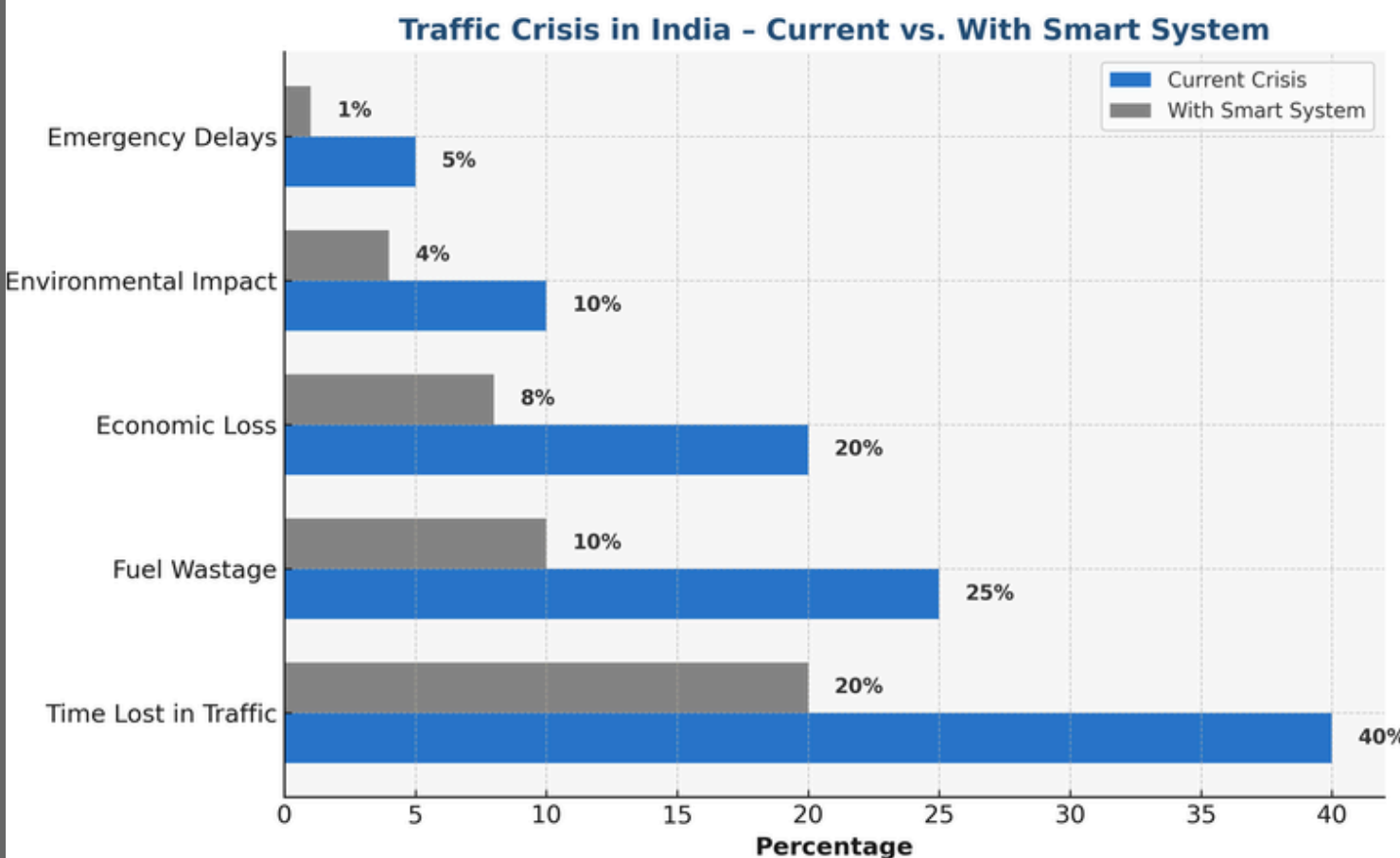
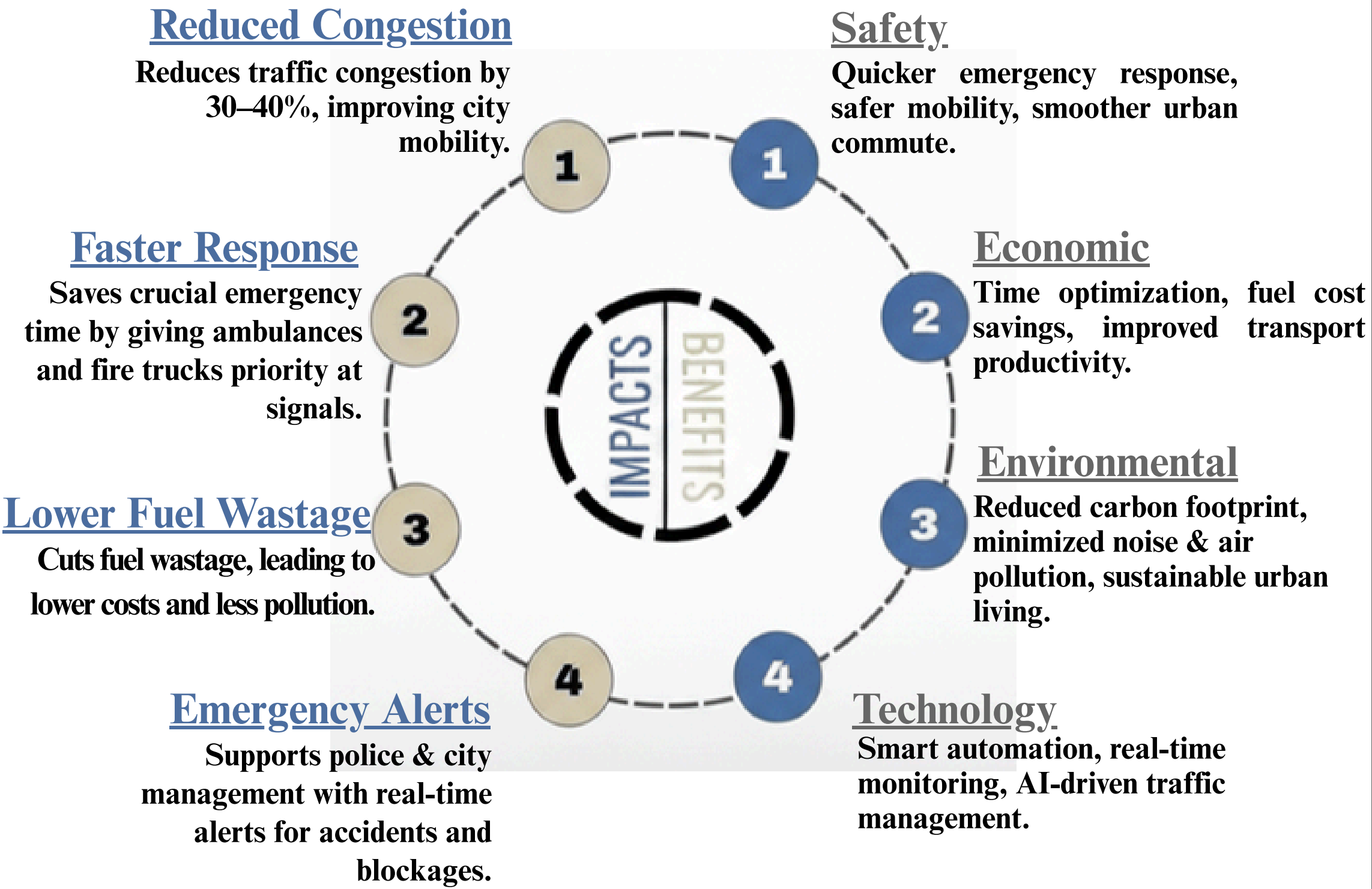


Government Cost Savings: Automates manual monitoring to significantly reduce operational costs and optimize the allocation of traffic personnel.

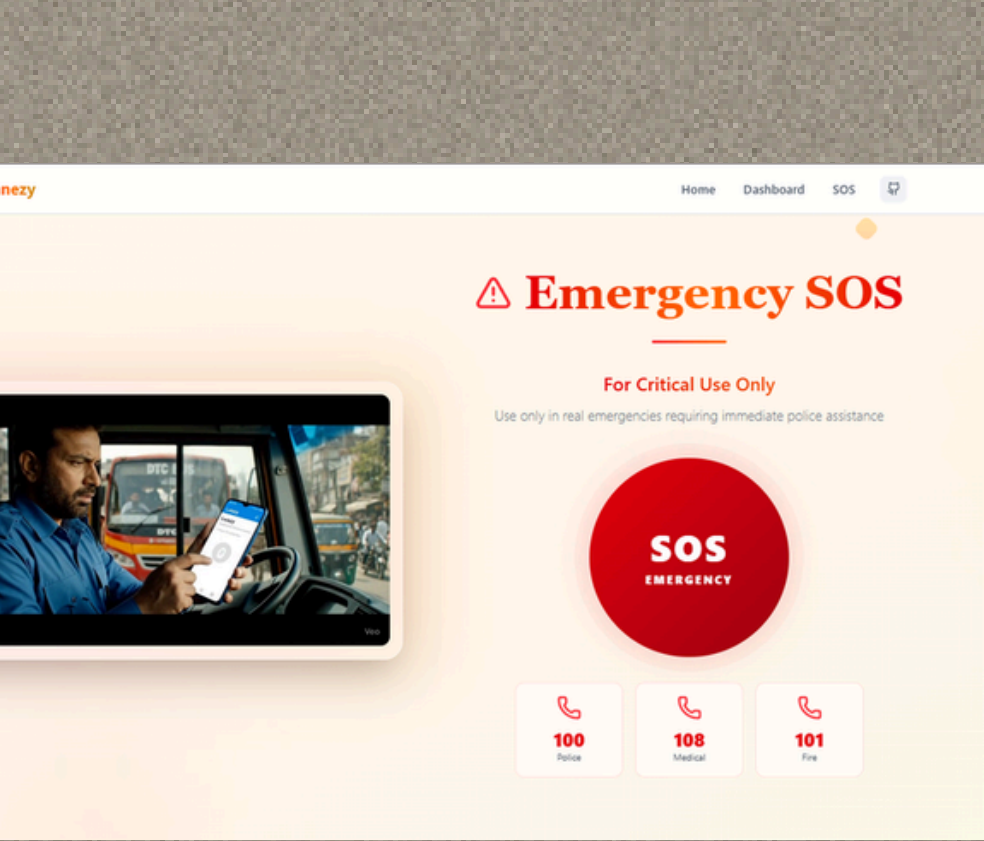
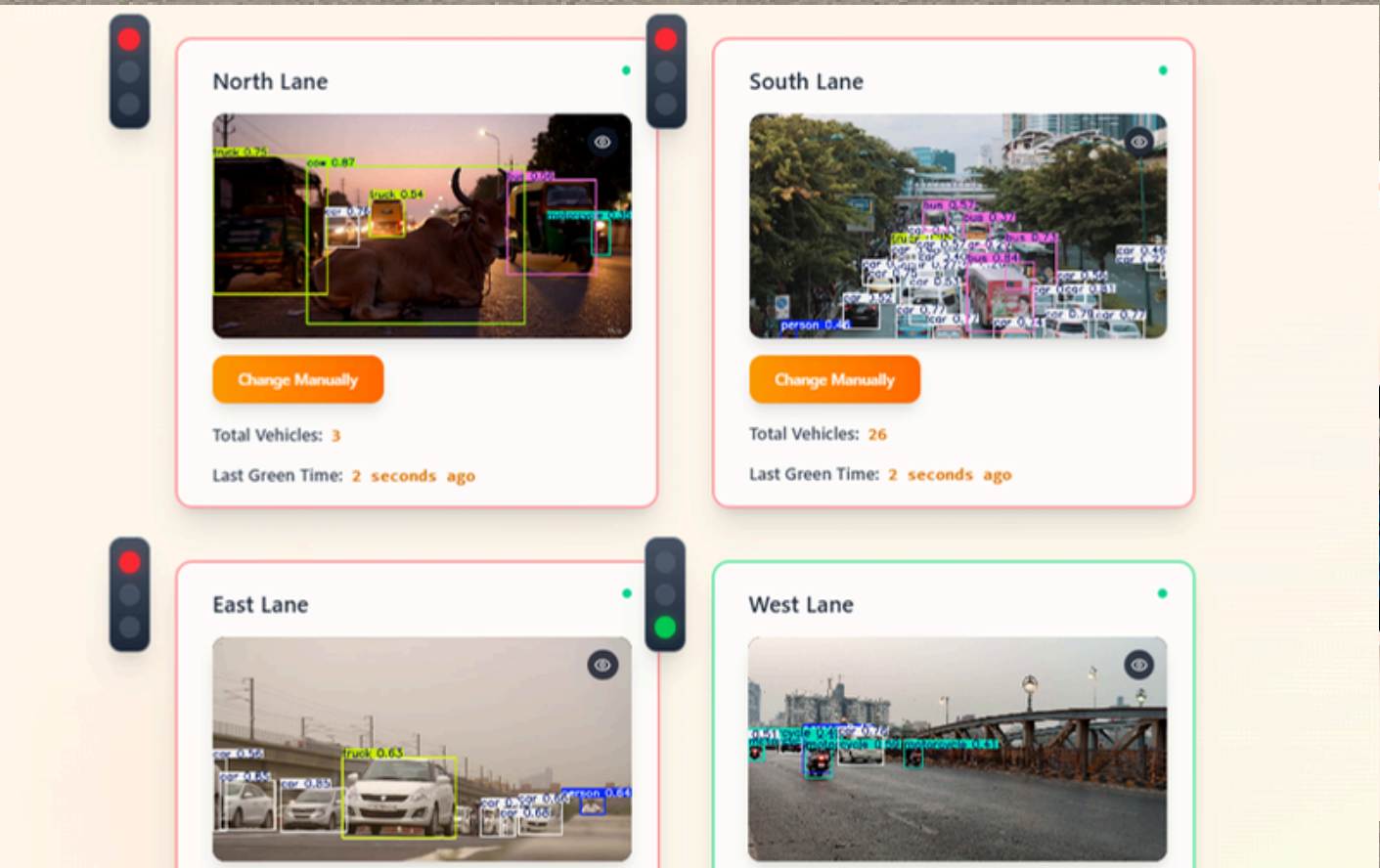
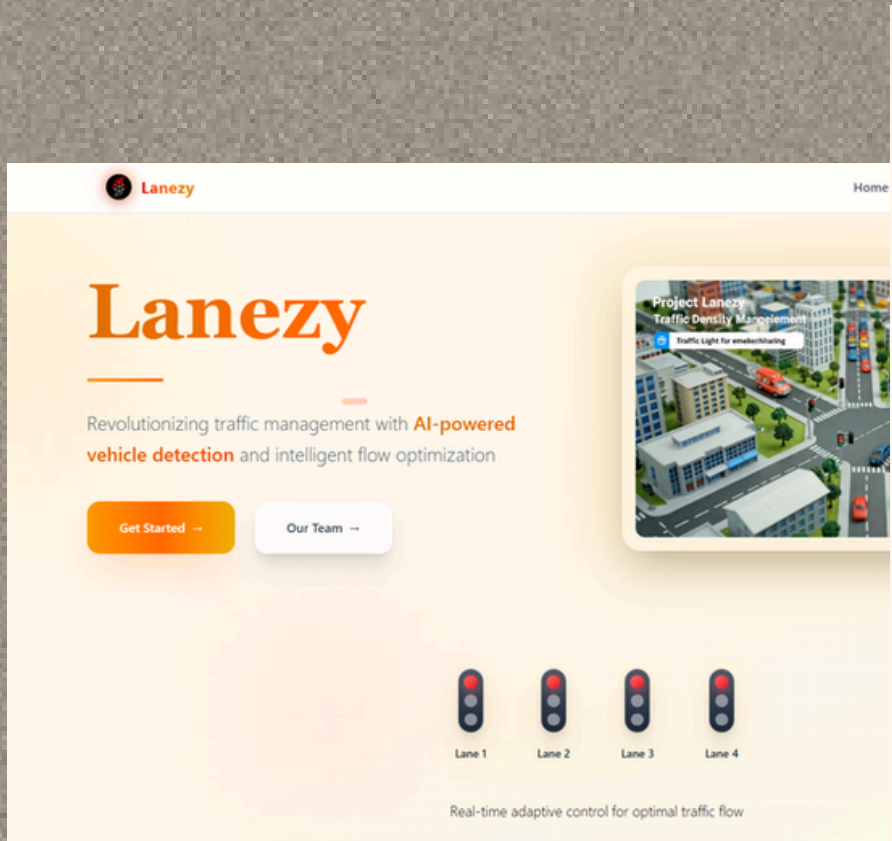
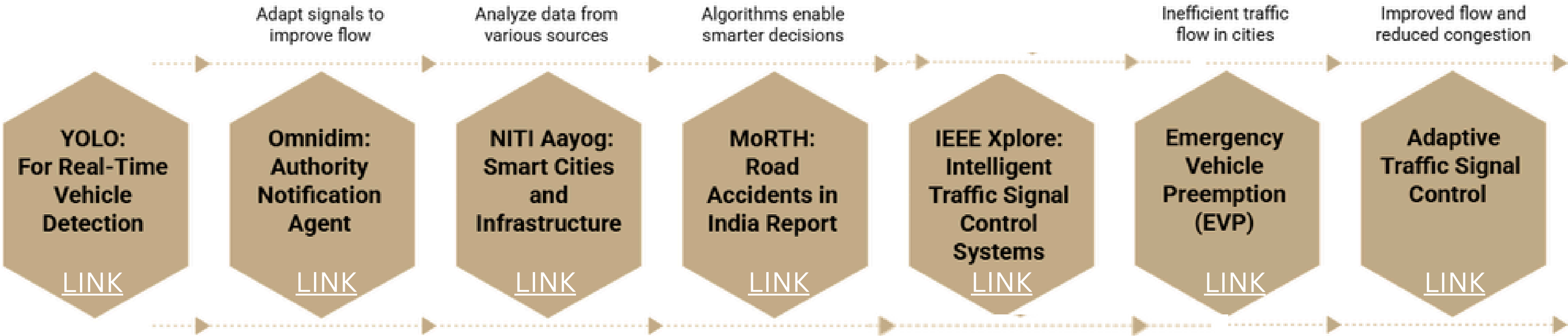


New Revenue Generation: Creates a direct and sustainable revenue stream through seamless integration with e-challan systems for automated fine collection.





RESEARCH AND REFERENCES



UI/UX